Emre Can Kadirhan

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SKILLS

Programming Languages: Python**,** Java, JavaScript, TypeScript, C++
Frameworks: FastAPI, Django, React, Spring, Spring Boot, Next.js
Tools and Technical Skills: Git/GitHub, Docker, CI/CD, Linux/Unix, Postman, PyTest, JUnit, PostgreSQL, MySQL, Prisma, SQLAlchemy, Redis, Celery

EXPERIENCE

Software Engineer Intern at TQueue Software ***June 2024 – August 2024***

- Developed and maintained RESTful APIs using Spring Boot for a customer queue management system that served as the primary API layer for the core product
- Refactored the customer management module into a more service-oriented architecture, reducing technical debt
- Wrote unit and integration tests using JUnit, increasing test coverage on the customer service module from near zero to roughly 70%

PROJECTS

Bot Battle Arena | Python, FastAPI, Celery, Redis, PostgreSQL, SQLAlchemy, Docker | GitHub

- Engineered a referee system that builds Docker containers to execute untrusted bot code in fully isolated sandboxes with disabled networking, memory caps, PID limits, and dropped Linux capabilities
- Built an async match execution pipeline using Celery and Redis, fully decoupling match orchestration from the API layer to prevent container workloads blocking web requests
- Implemented real-time match streaming using FastAPI WebSockets and Redis pub/sub, bridging the Celery worker and API server processes to broadcast live game state to connected viewers
- Designed a PostgreSQL schema with turn-by-turn state snapshots enabling complete match replay

SwingVision - AI Golf Swing Analyzer | Next.js, Python, PostgreSQL, Redis, MediaPipe, BetterAuth, Prisma| GitHub

- Built a Python worker using OpenCV and MediaPipe to overlay real-time AI pose skeletons on user uploaded golf swing videos to analyze swing mechanics
- Integrated Redis based async task queue to offload video processing, keeping web response times under 200ms during heavy load
- Architected a PostgreSQL (Neon) schema to manage user accounts, video metadata, and professional coaching feedback with relational integrity
- Implemented role-based access control using Next.js and BetterAuth, supporting separate user and admin dashboard experiences

Rate Limiter Library | Python, FastAPI, Django, Redis | GitHub

- Designed and developed a reusable rate limiting library for FastAPI and Django with three configurable algorithms (Fixed Window, Sliding Window, Token Bucket) using Redis as a backend with CI/CD pipeline
- Architected a flexible API supporting both route-level decorators and global middleware, with a test suite validated against a live Redis instance via Docker

EDUCATION

Arizona State University – Tempe, AZ | B.S. in Computer Science | May 2027 | GPA: 3.86